Modeling Immune Activation & Self-Tolerance

T-cell, Treg, and IL-2 Dynamics

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https://gabrielcirker.github.io/math-463-miniproject/

Introduction

The immune system is a complex network "designed" to protect an individual from pathogens and is generally split into two categories: innate and adaptive. The innate immune system is inherent and does not form antigen-specific "memory" of encountered pathogens. The adaptive immune system, which this paper focuses on, develops over time as the body encounters new pathogens. After an initial encounter, it learns to recognize and handle the previously unfamiliar pathogen. It should react against nonself pathogens while remaining tolerant to self. However, some nonself pathogens can resemble self-antigens, making them harder to recognize and fight off. Additionally, the immune system must also handle "altered self," such as cancer cells.

One way the immune system deals with these disturbances is by producing T-cell clones with random specificities to recognize peptides from self to non-self and everything in between. Although T cells can recognize self-peptides, they should not react harmfully to them, and those that do are meant to undergo clonal deletion. This can create gaps in the immune system because it may hinder the ability to detect and handle self-similar nonself and altered-self disturbances. However, clonal deletion does not eliminate all autoreactive T cells, and those that escape can contribute to autoimmunity. Regulatory T cells (Tregs) partly control these escaped autoreactive cells. Natural Tregs make up the majority of peripheral Tregs. Khailaie et al. (2013) describe their selection from autoreactive T cells in the thymus as a hypothesis. Activated conventional T cells produce interleukin-2, a cytokine that supports the proliferation of both conventional T cells and Tregs. Excessive or inappropriate Treg activity, though, might prevent beneficial effector responses, as their main role is to regulate the immune system by suppressing effector responses.

Given this context, the paper asks: "How does the immune system balance the tolerance mechanisms in order to ensure immune responses to any kind of disturbances, including self-disturbances, yet staying tolerant to self in healthy homeostasis?" (Khailaie et al., 2013). To answer this question, the authors used a series of mathematical models, which we analyze in more detail later in this report. The models suggest that self-reactive T cells escaping deletion are not just an error, but that the immune system allows self-reactivity to some degree because it can be beneficial. Self-tolerance is maintained by limiting the renewal rate of low-avidity autoreactive T cells (cells that weakly recognize self-antigens and require stronger stimulation to activate) and ensuring a reasonably high renewal of Tregs. This helps prevent physiological self-antigen exposure from triggering an immune response but still allows responses to disturbances that provide enough stimulation. Maintaining the balance matters because inadequate regulation can lead to autoimmunity, but excessive regulation can suppress necessary or beneficial immune responses.

Immune Activation Model

To examine the balance between immune activation and regulation described above, we focus on a model from Khailaie et al. (2013) that represents the interaction between conventional T cells, Tregs, and IL-2. The model helps describe how a response can develop while regulatory activity limits the activation of conventional T cells.

We use an adapted three-state version of Eq. (13) from Khailaie et al. (2013). The variables represent activated conventional T cells (T), activated regulatory T cells (R), and IL-2 (I). In the original model, activation depends on the two resting-cell populations. In our version, we replace those activation terms with fixed inputs k and λk, where λ is the ratio of the regulatory T-cell input to the conventional T-cell input. This gives the following system:

$$I. \quad \frac{dT}{dt} = k + aIT - bT - cT^2 - \gamma RT$$

$$II. \quad \frac{dR}{dt} = \lambda k + \varepsilon aIR - bR$$

$$III. \quad \frac{dI}{dt} = dT - eI(T + R) - fI$$

Eq. (13a) describes the rate of change in activated conventional T cells over time. It can be read, through biological interpretations of each term, as the rate at which conventional T cells become activated due to stimulation by an antigen plus the IL-2-dependent proliferation of activated conventional T cells minus the number of activated conventional T cells that die naturally, the number of activated conventional T cells that die due to other activated conventional T cells, and the number of activated conventional T cells that are suppressed by Tregs (not necessarily killed).

Eq. (13.b) describes the rate of change in activated regulatory T cells, or Tregs, over time. It can be interpreted biologically as the rate at which Tregs become activated due to stimulation by an antigen plus the IL-2-dependent proliferation of activated Tregs minus the number of activated Tregs that die naturally (this death rate is assumed to be the same as that of the activated conventional T cells).

Eq. (13.c) describes the rate of change in IL-2 over time. It can be interpreted biologically as IL-2 secretion by activated conventional T cells (d refers to the rate at which activated conventional T cells produce IL-2, and T is the number of cells producing it) minus the consumption of IL-2 by both conventional T cells and Tregs (assumed to be consumed at the same rate by both although total consumption differs based on population sizes), and IL-2 loss (independent of its consumption).

Parameters, Numerical Setup, and Baseline Comparison

Table 1. Baseline parameters. Cell interaction and IL-2 parameters follow Khailaie et al. (2013, Table 1).

Symbol	Value	Role	Symbol	Value	Role
a	0.4	IL-2-driven T cell growth	d	0.01	IL-2 production by T cells
b	0.1	Natural loss of T cells and Tregs	e	0.01	IL-2 consumption by both populations
c	1×10^{-5}	Quadratic T cell loss	f	1	IL-2 decay
γ	0.1	Treg suppression of T cells	ε	0.6	Relative IL-2-driven growth of Tregs
k	1.4	Conventional T cell input	λ	0.006	Ratio of Treg to T cell input

The baseline has $k = 1.4$ and $\lambda = 0.006$. We start at $T = R = I = 0$ and integrate for 600 model time units. We use the Radau method in SciPy's *solve_ivp*, with relative and absolute tolerances of 10^{-8} and 10^{-10}. The maximum internal step is 0.5, and we save a sample every 0.05 model time units (Virtanen et al., 2020).

To check the numerical solution, we reran the baseline with relative tolerance 10^{-10}, absolute tolerance 10^{-12}, and a maximum internal step of 0.25. The largest difference after dividing by each state's range was $3.92 \cdot 10^{-9}$. That difference is small compared with the changes we report below.

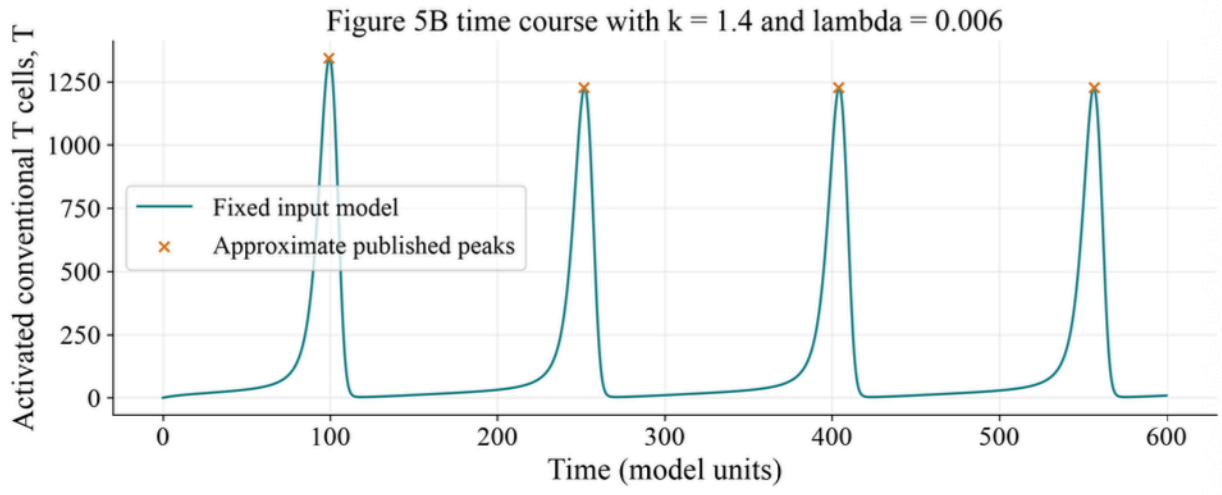


Figure 1. Simulated conventional T cell trajectory and four approximate peaks digitized from the middle inset of Figure 5B in Khailaie et al. (2013). The markers show readings from the published image.

The four peak times differed by at most *0.63* model time units, and the heights differed by at most *1.09* T cells. Comparing the full baseline trajectory with the width of the printed curve, *99.65%* of simulated values at *1,134* visible image columns fell inside the plotted stroke. This comparison measures agreement with the published inset at its image resolution. The authors' original numerical data were unavailable for the comparison.

Response Across the Figure 5 Settings

We also ran the other five parameter pairs shown in the time-course insets of Figure 5. We used $k = 0.2$, 1.4, and 8.5 at $\lambda = 0.020$ and 0.006. Each run used the same initial values, solver settings, and sampling interval as the baseline. We prescribed these inputs independently of the paper's resting pool parameters.

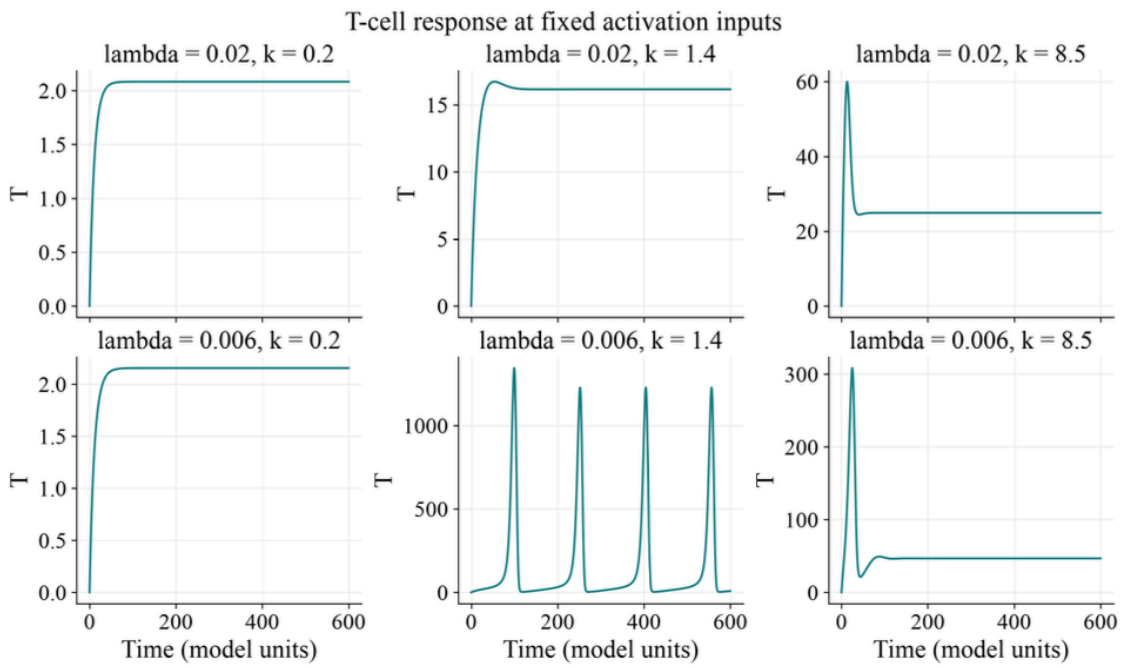


Figure 2. Activated conventional T cells over time at the six Figure 5 inset settings. The panels use different vertical scales so the smaller responses remain visible. Only the middle panel of the bottom row was checked against a digitized trace.

At $\lambda = 0.006$, the lowest input gives a small response. At the middle input, conventional T cells rise in repeated bursts. With the highest input, the first spike is followed by a much smaller response. Increasing k supplies more cells to both activated populations. At high input, enough regulatory cells can accumulate to suppress the conventional T cell response even though more conventional cells are entering the population.

At $\lambda = 0.020$, all three curves settle near steady levels well below the baseline peaks. This comparison gives us a reason to hold k fixed and change λ by itself. The six panels show the time courses from the fixed-input model. The locations of bifurcations in the paper's Figure 5 come from a separate stability analysis.

Changing the Regulator Cell Input

We held k at *1.4* and compared λ = *0.006*, *0.009*, *0.012*, and *0.020*. This changes the number of newly activated Tregs entering the model per time unit while leaving the conventional T cell input and all other parameters fixed.

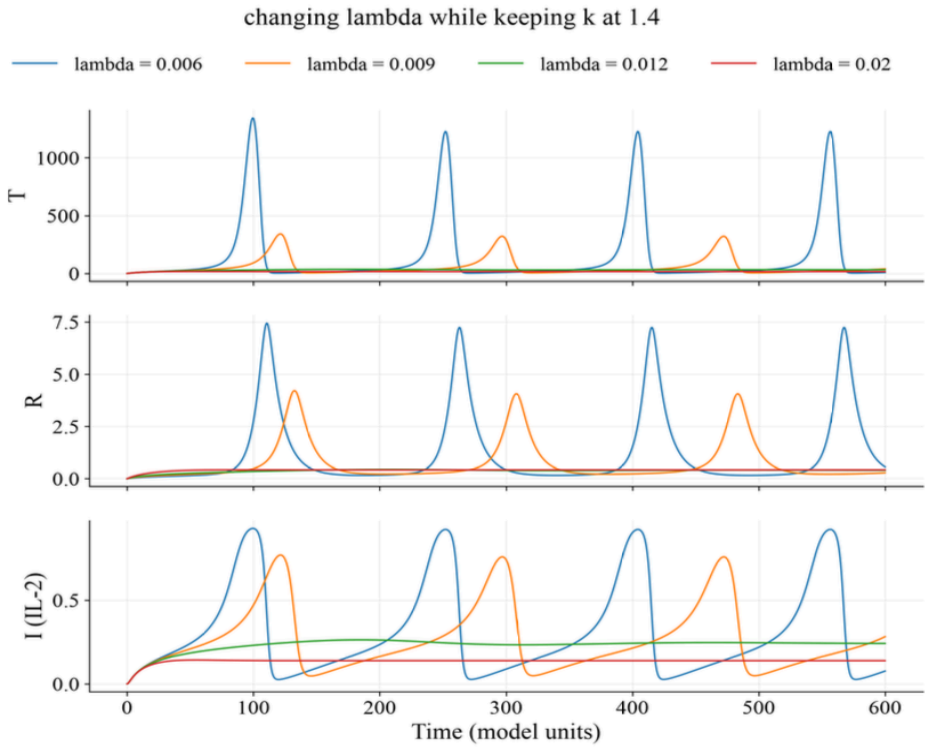


Figure 3. Conventional T cells (T), regulatory T cells (R), and IL-2 (I) at four values of λ. Increasing λ reduces the large bursts in T over the observed window.

To estimate the late peak spacing, we used peaks from the final 75% of each run. A peak had to have a prominence of at least 5% of the maximum absolute T value in that window, or 10^{-6}, whichever was larger. We calculated the median spacing when at least two peaks were detected.

Table 2. Maximum T over the full run and median peak spacing in the final *75%* of the run.

λ	Maximum T	Late peak spacing	Intervals used
0.006	1343.397	152.225	2
0.009	341.948	175.350	1
0.012	35.851	Not detected	0
0.020	16.718	Not detected	0

The peak at λ = *0.009* is about one quarter of the baseline peak, and the late bursts are farther apart. That spacing comes from one interval. The two larger λ values had too few qualifying late peaks to estimate a spacing. More Tregs enter directly through λk, so suppression depends less on the build-up of IL-2 from a large conventional T cell response. The finite time window only tells us what was observed in these runs.

Learning the Equations with SINDy

We used SINDy to see whether simulated measurements of T, R, and I could recover the terms in the model (Brunton et al., 2016). For each state, we used a fourth-order centered difference to estimate its time derivative. We dropped two points at each end of the training window so every derivative estimate used only samples from that window.

The candidate terms were 1, T, R, I, T^2, TR, TI, R^2, RI, and I^2. We scaled the states, candidate terms, and derivative estimates using training data. Sequentially thresholded least squares first fits the candidate terms to the derivatives, then removes terms with small coefficients and refits the remaining terms. We converted the learned coefficients back to the original units before simulating the learned equations.

Table 3. Data used for fitting, model selection, and testing. All times use model units.

Data	Initial state (T, R, I)	Duration	Use
Training	(0, 0, 0)	50, 150, or 300	Fit the coefficients
Validation	(25, 0.25, 0.10)	120	Choose the threshold
Test	(50, 0.50, 0.15)	600	Compare trajectories

For each threshold, we compare the estimated rates of change with the rates predicted by the learned equations on the validation trajectory. We divided each derivative mean-square error by the mean square of its validation derivative, averaged the results across the three states, and added 10^{-5} per retained term. We chose the lowest-scoring fit that completed a nonnegative validation trajectory. The validation data chose the threshold, and the test trajectory was kept separate until the final comparison.

Table 4. Recovery of the 12 true terms and test errors. NRMSE is the root-mean-square trajectory error divided by the range of the corresponding true test state.

λ	Train time	True terms	T NRMSE	R NRMSE	I NRMSE
0.006	50	12/12	2.55E-06	1.91E-06	1.23E-06
0.006	150	12/12	2.11E-07	1.63E-07	1.81E-07
0.006	300	12/12	1.84E-07	1.42E-07	1.57E-07
0.020	300	11/12	6.72E-04	9.79E-04	6.27E-04

All four learned systems completed the 600-unit test and stayed nonnegative within the solver check tolerance. At $\lambda = 0.006$, the 50-unit window already recovered all *12* terms. Training for 150 or 300 units reduced the test errors further without changing which terms were selected. The 50-unit window contains *1,001* saved samples and still uses the separate 120-unit validation trajectory to choose the threshold.

Comparing the Learned Trajectories and Equations

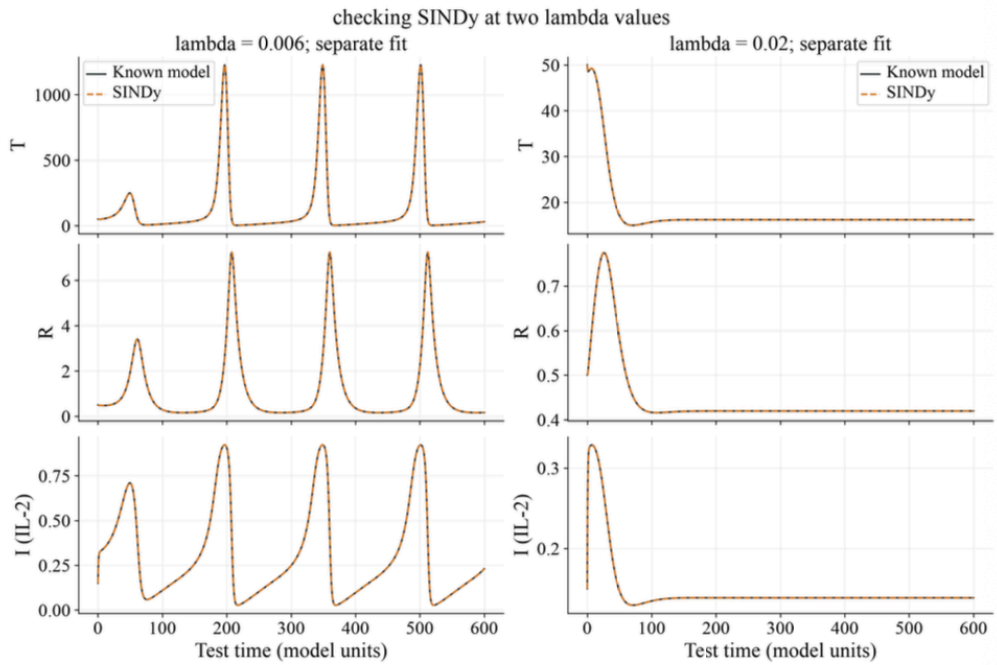


Figure 4. Test trajectories from the known model and the learned equations after 300 model time units of training. The two columns use separate fits at $\lambda = 0.006$ and $\lambda = 0.020$. Each starts from the held-out initial values in Table 3. The dashed SINDy curves nearly cover the solid known-model curves.

At $\lambda = 0.006$, the learned model recovered all *12* terms with no extra terms. For the 300-unit run, the learned coefficient of T^2 was $-1.00000015 \cdot 10^{-5}$, compared with $-1.0 \cdot 10^{-5}$ in the known model. At $\lambda = 0.020$, the fit kept *11* of the original terms and dropped the quadratic loss term in the T equation. The equations below compare the known and learned rates of change for T in that case.

Known $\quad \frac{dT}{dt} = 1.4 - 0.1T - 10^{-5}T^2 - 0.1TR + 0.4TI$

Learned $\quad \frac{dT}{dt} \approx 1.399954 - 0.099990T - 0.099992TR + 0.398755TI$

The learned R and I equations retained all their original terms in the quiet case. Even with the missing term, the T test error was about *0.0672%* of its range. During the 300-unit training runs, the quadratic loss term reached a magnitude of about *18* at baseline and only *0.0028* in the quiet case. The small contribution helps explain why the fit could drop the term and still match the test curves.

Discussion

The parameter sweep helps explain why the amount of regulation matters in the paper's model. At fixed conventional T cell input, raising the regulatory input reduced the peak response from about *1,343* to about *17* cells across the tested settings. The equations give a reason for this pattern. More Tregs enter the system even when IL-2 is low, and as the regulatory population builds, it can suppress conventional T cells before they reach the peaks seen at baseline.

The SINDy comparison shows a consequence of that smaller response. The baseline trajectory covers both low and high values of T, so it contains information about the model in a wider range of states. The quiet trajectory gives good predictions for the test initial condition even without the quadratic term. We would need more varied initial conditions or data with larger excursions to find out whether those observations help recover the missing term.

The model comparison has two limits. First, the fixed inputs replace the paper's changing resting cell pools, so our baseline check speaks to one published trajectory and its plotted precision. Second, SINDy used clean simulated data and a quadratic library that already contained the true terms. The results show what the method can recover under those conditions. In particular, a good forecast and recovery of the underlying equations answer different questions.

AI Assistance

GitHub Copilot was used through Visual Studio Code to assist with debugging code executed in the integrated terminal. The authors reviewed and verified the suggested changes and are responsible for the final code and results.

Grammarly was used to proofread the manuscript and assist with corrections to grammar, spelling, and clarity. The authors reviewed the suggestions and made the final decisions about the writing.

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